CHAPTER 11

Disk & Secondary Storage

Disk geometry · capacity · access time = seek + rotation + transfer · transfer rate · IOPS

QUICK TRICK

Access time = seek + rotational latency + transfer time (+ controller overhead). Average rotational latency = **half a rotation = 30 / RPM seconds**. Capacity = surfaces × tracks × sectors × bytes per sector.

(The 2027 syllabus wording does not name secondary storage. These are short, easy marks if asked, so keep this page for revision.)

11.1 Disk structure

● **LEVEL 1 · BEGINNER**

A hard disk has one or more **platters** spinning together. Each platter surface has concentric **tracks**; each track is split into **sectors** (the smallest unit read/written, often 512 B or 4 KB). The same track number on all surfaces forms a **cylinder**. A head per surface moves in and out together.

Analogy. A record player: moving the needle to the right groove (seek), waiting for the right spot to come around (rotation), then playing it (transfer).

11.2 Capacity

● **LEVEL 2 · EXAM**

```
Capacity = (surfaces) x (tracks per surface) x (sectors per track) x (bytes per sector)
Cylinders = tracks per surface        Surfaces = 2 x platters (if both sides used)
```

Example. 8 surfaces, 1000 tracks, 200 sectors per track, 512 B per sector: 8 × 1000 × 200 × 512 = **819,200,000 B** (about 819 MB, 781 MiB).

11.3 Access time

● **LEVEL 2 · EXAM**

Component	Meaning	Formula
Seek time	Move head to the correct track	given (often "average seek")
Rotational latency	Wait for the sector to arrive under the head	average = ½ × (60 / RPM)
Transfer time	Read the data	bytes ÷ transfer rate, or (sectors ÷ sectors per track) × rotation time
Controller overhead	Electronics	given

- **Rotation time** = 60 / RPM seconds. 7200 RPM → 8.33 ms; average latency = **4.17 ms**. 15,000 RPM → 4 ms; latency = **2 ms**.
- **Transfer rate** = (sectors per track × bytes per sector) ÷ rotation time. 200 sectors × 512 B at 6000 RPM (10 ms per rotation) → 102,400 B / 10 ms = **10.24 MB/s**.

Example. Average seek 6 ms, 7200 RPM, 100 sectors per track, read one sector: latency 4.17 ms, transfer = 8.33 ms / 100 = 0.083 ms. Access time = 6 + 4.167 + 0.083 = **10.25 ms**.

Example. 4 KB block (8 sectors of 512 B) on a track of 100 sectors at 7200 RPM, seek 9 ms: transfer = 8/100 × 8.33 = 0.67 ms. Total = 9 + 4.17 + 0.67 = **13.84 ms**, so about 1/0.01384 ≈ **72 random I/Os per second**.

QUICK TRICK

Random access is dominated by **seek + rotation**; sequential access mostly pays the transfer time. That is why contiguous layout is fast.

11.4 HDD vs SSD

● LEVEL 1 · BEGINNER

	HDD	SSD
Mechanism	Spinning platters, moving head	Flash memory, no moving parts
Random access	Slow (ms)	Fast (tens of µs)
Cost per GB	Lower	Higher
Wear	Mechanical	Limited write cycles

11.5 Common mistakes

COMMON MISTAKE

- Using a full rotation instead of half for the **average** rotational latency.
- Forgetting that transfer time depends on the fraction of the track read.
- Forgetting to double the platter count to get surfaces.
- Mixing ms and µs.

GATE QUICK CHECK — Chapter 11

```
access = seek + 0.5*(60/RPM) + transfer (+overhead)        rotation = 60/RPM
transfer rate = (sectors*bytes)/rotation time              capacity = surf*tracks*sect*bytes
7200 RPM -> latency 4.17 ms | 15000 RPM -> 2 ms
```

Practice questions

● LEVEL 1 · BEGINNER

Q1. Average rotational latency at 7200 RPM.

Q2. Average rotational latency at 15,000 RPM.

Q3. What is a cylinder?

Q4. Which part dominates random access time on an HDD?

● LEVEL 2 · EXAM

Q5. Capacity: 8 surfaces, 1000 tracks, 200 sectors per track, 512 B per sector.

Q6. 200 sectors of 512 B per track, 6000 RPM. Transfer rate?

Q7. Seek 6 ms, 7200 RPM, 100 sectors per track. Time to read one sector?

Q8. How many surfaces does a 2-platter disk (both sides used) have?

● LEVEL 3 · GATE

Q9. 4 KB block (8 sectors) on a 100-sector track, 7200 RPM, seek 9 ms. Access time and random IOPS?

Q10. 2 platters (4 surfaces), 10,000 tracks per surface, 500 sectors per track, 512 B sectors. Capacity?

Q11. Why is reading a large file stored contiguously faster than the same file scattered over the disk?

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Answers

Q	Solution
1	0.5 × 8.33 = **4.17 ms**.
2	0.5 × 4 ms = **2 ms**.
3	The set of tracks with the same track number on all surfaces.
4	**Seek and rotational latency**.
5	8 × 1000 × 200 × 512 = **819,200,000 B**.
6	102,400 B per 10 ms = **10.24 MB/s**.
7	6 + 4.167 + 0.083 = **10.25 ms**.
8	**4**.
9	Transfer 0.67 ms. Total 9 + 4.17 + 0.67 = **13.84 ms** → ≈ **72 IOPS**.
10	4 × 10,000 × 500 × 512 = **10.24×10^9 B** (10.24 GB).
11	Contiguous data needs one seek and one rotational wait, then streams at the transfer rate; scattered data pays seek and rotation for every piece.

Next: Chapter 12 — I/O: Interrupt and DMA.

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